

RESEARCH SYNTHESIS

STATISTICAL APPROACHES TO PROTECTING CONFIDENTIALITY FOR MICRODATA AND THEIR EFFECTS ON THE QUALITY OF STATISTICAL INFERENCES

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Abstract When sharing microdata (i.e., data on individuals) with the public, organizations face competing objectives. On the one hand, they strive to release data files that are useful for a wide range of statistical purposes and easy for secondary data users to analyze with standard statistical methods. On the other hand, they must protect the confidentiality of data subjects' identities and sensitive attributes from attacks by ill-intentioned data users. To address such threats, organizations typically alter microdata before sharing it with others; in fact, most public-use data sets with unrestricted access have undergone one or more disclosure protection treatments. This research synthesis reviews statistical approaches to protecting data confidentiality commonly used by government agencies and survey organizations, with particular emphasis on their impacts on the accuracy of secondary data analyses. In general terms, it discusses potential biases that can result from the disclosure treatments, as well as when and how it is possible to avoid or correct those biases. The synthesis is intended for social scientists doing secondary data analysis with microdata; it does not prescribe best practices for implementing disclosure protection methods or gauging disclosure risks when disseminating microdata.

Introduction

Many national statistical agencies, survey organizations, and research centers—henceforth all called agencies—disseminate microdata, i.e., data on individual records, to the public. Wide dissemination of microdata facilitates advances

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in social science and public policy, helps citizens learn about their communities, and enables students to develop skills at data analysis. Wide dissemination enables others to avoid mounting unnecessary new surveys when existing data suffice to answer questions of interest, and it helps agencies improve the quality of future data collection efforts via feedback from those analyzing the data. Finally, wide dissemination provides funders of the survey, e.g., taxpayers, with access to what they paid for.

Often, however, agencies cannot release microdata as collected because doing so could reveal survey respondents' identities or values of sensitive attributes. Failure to protect confidentiality can have serious consequences for the agency, since it may violate laws passed to protect confidentiality, such as the Health Insurance Portability and Accountability Act and the Confidential Information Protection and Statistical Efficiency Act in the United States. Additionally, when confidentiality is compromised, the agency may lose the trust of the public, so that potential respondents are less willing to give accurate answers, or even to participate, in future surveys (Reiter 2004).

At first glance, sharing safe microdata seems a straightforward task: simply strip unique identifiers like names, addresses, and tax identification numbers before releasing the data. However, anonymizing actions alone may not suffice when other readily available variables, such as aggregated geographic or demographic data, remain on the file. These quasi-identifiers can be used to match units in the released data to other databases. For example, Sweeney (2001) showed that 97 percent of the records in publicly available voter registration lists for Cambridge, Massachusetts, could be uniquely identified using birth date and nine-digit zip code. By matching on the information in these lists, she was able to identify Governor William Weld in an anonymized medical database. More recently, the company Netflix released supposedly de-identified data describing more than 480,000 customers' movie viewing habits; however, Narayanan and Shmatikov (2008) were able to identify several customers by linking to an online movie ratings website, thereby uncovering apparent political preferences and other potentially sensitive information.

Although these re-identification exercises were done by academics to illustrate concerns over privacy, one easily can conceive of re-identification attacks for nefarious purposes, especially for large social science databases. A nosy neighbor or relative might search through a public database in an attempt to learn sensitive information about someone they knew had participated in a survey. A journalist might try to identify politicians or celebrities. Marketers or creditors might mine large databases to identify good, or poor, potential customers. And, disgruntled hackers might try to discredit organizations by identifying individuals in public-use data.

It is difficult to quantify the likelihood of these scenarios playing out; agencies generally do not publicly report actual breaches of confidentiality, and it is not clear that they would ever learn of successful attacks. Nonetheless, the threats alone force agencies to react. For example, the national statistical system

would be in serious trouble if a publicized breach of confidentiality caused response rates to nosedive. Agencies therefore further limit what they release by altering the collected data. These methods can be applied with varying intensities. Generally, increasing the amount of data alteration decreases the risks of disclosures, but it also decreases the accuracy of inferences obtained from the released data, since these methods distort relationships among the variables (Duncan, Keller-McNulty, and Stokes 2001).

Typically, analysts of public-use data do not account for the fact that the data have been altered to protect confidentiality. Essentially, secondary data analysts act as if the released data are in fact the collected sample, thereby ignoring any inaccuracies that might result from the disclosure treatment. This is understandable default behavior. Descriptions of disclosure protection procedures are often vague and buried within survey design documentation. Even when agencies release detailed information about the disclosure-protection methods, it can be non-trivial to adjust inferences to properly account for the data alterations. Nonetheless, secondary data analysts need to be cognizant of the potential limitations of public-use data.

In this article, we review the evidence from the literature about the impacts of common statistical disclosure limitation (SDL) techniques on the accuracy of statistical inferences. We document potential problems that analysts should know about when analyzing public-use data, and we discuss in broad terms how and when analysts can avoid these problems. We do not synthesize the literature on implementing disclosure protection strategies; that topic merits a book-length treatment (see Willenborg and de Waal 2001, for instance). We focus on microdata and do not discuss tabular data, although many of the SDL methods presented here, and their corresponding effects on statistical inference, apply for tabular data.

The remainder of the article is organized as follows. Section 2 provides an overview of the context of data dissemination, broadly outlining the trade-offs between risks of disclosure and usefulness of data. Section 3 describes several confidentiality protection methods and their impacts on secondary analysis. Section 4 concludes with descriptions of some recent research in data dissemination, and speculates on the future of data access in the social sciences if trends toward severely limiting data releases continue.

Setting the Stage: Disclosure Risk and Data Usefulness

When making data sharing or dissemination policies, agencies have to consider trade-offs between disclosure risk and data usefulness. For example, one way to achieve zero risk of disclosures is to release completely useless data (e.g., a file of randomly generated numbers) or not to release any data at all; one way to achieve high usefulness is to release the original data without any concern for confidentiality. Neither of these is workable; society in

general accepts some disclosure risks for the benefits of data access. This trade-off is specific to the data at hand. There are settings in which accepting slightly more disclosure risks leads to large gains in usefulness, and others in which accepting slightly lower data quality leads to great reductions in disclosure risks.

To make informed decisions about the trade-off, agencies generally seek to quantify disclosure risk and data usefulness. For example, when two competing SDL procedures result in approximately the same disclosure risk, the agency can select the one with higher data usefulness. Additionally, quantifiable metrics can help agencies decide if the risks are sufficiently low, and the usefulness adequately high, to justify releasing the altered data.

In this section, we present an overview of disclosure risk and data usefulness quantification.¹ The intention is to provide a context in which to discuss common SDL procedures and their impacts on the quality of secondary data analyses. The overview does not explain how to implement disclosure risk assessments or data usefulness evaluations. These are complex and challenging tasks, and agencies have developed a diverse set of approaches for doing so.²

IDENTIFICATION DISCLOSURE RISK

Most agencies are concerned with preventing two types of disclosures, namely (1) identification disclosures, which occur when a malicious user of the data, henceforth called an intruder, correctly identifies individual records in the released data; and (2) attribute disclosures, which occur when an intruder learns the values of sensitive variables for individual records in the data. Attribute disclosures usually are preceded by identity disclosures—for example, when original values of attributes are released, intruders who correctly identify records learn the attribute values—so that agencies focus primarily on identification disclosure risk assessments. See the reports of the National Research Council (2005, 2007), Statistical Working Paper 22 (Federal Committee on Statistical Methodology 2005), and Lambert (1993) for more information about attribute disclosure risks.

Many agencies base identification disclosure risk measures on estimates of the probabilities that individuals can be identified in the released data. Probabilities of identification are easily interpreted: the larger the probability, the greater the risk. Agencies determine their own threshold for unsafe probabilities, and these typically are not made public.

1. Portions of the discussion in this section are drawn from the report of the National Research Council (2007, pp. 37–40).

2. For additional information on how agencies balance risk and utility, readers can investigate the website of the Committee on Privacy and Confidentiality of the American Statistical Association (www.amstat.org/committees/pc/index.html) and the references linked on that site.

There are two main approaches to estimating these probabilities. The first is to match records in the file being considered for release with records from external databases that intruders could use to attempt identifications (e.g., Domingo-Ferrer and Torra 2003; Paass 1988; Skinner 2008; Yancey, Winkler, and Creecy 2002). The matching is done by (1) searching for the records in the external database that look as similar as possible to the records in the file being considered for release; (2) computing the probabilities that these matching records correspond to records in the file being considered for release, based on the degrees of similarity between the matches and their targets; and (3) declaring the matches with probabilities exceeding a specified threshold as identifications. As a “worst case” analysis, the agency could presume that intruders know all values of the unaltered, confidential data, and match the candidate release file against the confidential file (Spruill 1982). This is easier and less expensive to implement than obtaining external data. For either case, agencies determine their own thresholds for unsafe numbers of correct matches and desirable numbers of incorrect matches.

The second approach is to specify conditional probability models that explicitly account for (1) assumptions about what intruders might know about the data subjects; and (2) any information released about the disclosure control methods. For the former, typical assumptions include whether or not the intruder knows that certain individuals participated in the survey, which quasi-identifying variables the intruder knows, and the amount of measurement error in the intruder’s data. For the latter, released information might include the percentage of swapped records or the magnitude of the variance when adding noise (see Section 3); it might also contain nothing but general statements about how the data have been altered when these parameters are kept secret. For illustrative computations of model-based identification probabilities, see Duncan and Lambert (1986, 1989), Fienberg, Makov, and Sanil (1997), Reiter (2005a), Drechsler and Reiter (2008), Hockett and Larsen (2008), and Shlomo and Skinner (2009).

Most agencies consider individuals who are unique in the population (as opposed to the sample) to be particularly at risk (Bethlehem, Keller, and Pannekoek 1990). Therefore, much research has gone into estimating the probability that a sample unique record is in fact a population unique record. Many agencies use a variant of Poisson regression to estimate these probabilities; see Elamir and Skinner (2006) and Skinner and Shlomo (2008) for reviews of this research.

A key issue in computing probabilities of identification, and in all disclosure risk assessments, is that the agency does not know what information unintentioned users have about the data subjects. Hence, agencies frequently examine risks under several scenarios, e.g., no knowledge versus complete knowledge of who participated in the study. By gauging the likelihood of those scenarios, the agency can determine if the data usefulness is high enough to be worth the risks under the different scenarios. This is an

imperfect process that is susceptible to miscalculation, e.g., the high-risk scenarios could be more likely than suspected. Such imperfections are arguably inevitable when agencies also seek to provide public access to quality microdata.

DATA USEFULNESS

Data usefulness is usually assessed with two general approaches: (1) comparing broad differences between the original and released data; and (2) comparing differences in specific models between the original and released data. Broad difference measures essentially quantify some statistical distance between the distributions of the data on the original and released files, for example, a Kullback-Leibler or Hellinger distance (Shlomo 2007). As the distance between the distributions grows, the overall quality of the released data generally drops. Computing statistical distances between multivariate distributions for mixed data is a difficult computational problem, particularly since the population distributions are not known in genuine settings. One possibility is to use *ad hoc* approaches, for example, measuring usefulness by a weighted average of the differences in the means, variances, and correlations in the original and released data, where the weights indicate the relative importance that those quantities are similar in the two files (Domingo-Ferrer and Torra 2001). Another strategy is based on how well one can discriminate between the original and altered data. For example, Woo et al. (2009) stack the original and altered data sets in one file, and estimate probabilities of being “assigned” to the original data conditional on all variables in the data set via logistic regression. When the distributions of probabilities are similar in the original and altered data, theory from propensity score matching—a technique commonly used in observational studies (Rosenbaum and Rubin 1983)—indicates that distributions of the variables are similar; hence, the altered data should have high utility.

Comparison of measures based on specific models is often done informally. For example, agencies look at the similarity of point estimates and standard errors of regression coefficients after fitting the same regression on the original data and on the data proposed for release. If the results are considered close, for example, the confidence intervals obtained from the models largely overlap, then the released data have high utility for that particular analysis (Karr et al. 2006). Such measures are closely tied to how the data are used, but they provide a limited representation of the overall quality of the released data. Thus, agencies examine models that represent the wide range of uses of the released data. For example, in the National Assessment of Adult Literacy, the National Center for Education Statistics evaluates the quality of proposed releases by comparing the estimates from the observed data and disclosure-proofed data for a large number of pairwise correlations and regression coefficients (Dohrmann et al. 2009).

Confidentiality Protection Methods and Their Impacts on Secondary Analysis

In this section, we describe several common strategies for confidentiality protection used by both government agencies and private organizations, including recoding variables, swapping data values for selected records, adding noise to variables, and creating partially synthetic data, i.e., replacing sensitive information with values simulated from statistical models. We outline in broad terms how agencies implement each approach, and we present findings from the literature on the impacts of these approaches on the accuracy of statistical inferences. We note that multiple methods can be employed on any one file and sometimes even on any one variable; see Oganian and Karr (2006) for an example of the latter.

RECODING

The basic idea of recoding is to use deterministic rules to coarsen the original data before releasing them. Examples include releasing geography as aggregated regions; reporting exact values only below or above certain thresholds (known as top- or bottom-coding), such as reporting all incomes above \$100,000 as “\$100,000 or more” or all ages above 90 as “90 or more”; and collapsing levels of categorical variables into fewer levels, such as detailed occupation codes into broader categories. Recoding reduces disclosure risks by turning atypical records—which generally are most at risk—into typical records. For example, there may be only one person with a particular combination of demographic characteristics in a county but many people with those characteristics in a state. Releasing data for this person with geography at the county level might have a high disclosure risk, whereas releasing the data at the state level might not. As this example suggests, agencies specify recodes to reduce estimated probabilities of identification to acceptable levels, e.g., to ensure that no cell in a cross-tabulation of a set of key categorical variables has fewer than k individuals, where k often is three or five.

Recoding is the most commonly used approach to protecting confidentiality. For example, the U.S. Census Bureau typically does not release geographical detail below 100,000 people in its public-use data products. The Census Bureau also uses top-coding to protect large monetary values in the American Community Survey and Current Population Survey public-use samples. The Health and Retirement Survey aggregates geographies, recodes occupation into coarse categories, and employs top-coding to protect monetary data. The American National Elections Studies (ANES) uses recoding as the primary protection scheme for the public-use file: it is created by aggregating geography to high levels and coarsening occupation detail from dozens of categories down to 20 or fewer. As these examples indicate, often multiple recodes are used simultaneously; in fact, the safe harbor confidentiality rules in the Health Insurance

Portability and Accountability Act—which governs health-related data—require aggregation of geography to units of no less than 20,000 individuals and top-coding of all ages above 90.

How does recoding affect the accuracy of secondary data analyses? Analyses intended to be consistent with the recoding are unaffected. For example, aggregation of geography to large regions has no impact on national estimates. Similarly, collapsing categories is immaterial to analyses that would use the recoded variables regardless. However, recoding has deleterious effects on other analyses. Aggregation of geography can create ecological fallacies, which arise when relationships estimated at the high level of geography do not apply at lower levels of geography. It smoothes spatial variation, which causes problems for both point and variance estimation when estimates differ by region. Collapsing categories masks heterogeneity among the collapsed levels; e.g., lumping legislators, financial managers, and funeral directors together (as is done in the ANES occupation recoding) into a common code hides differences among these types of individuals. Top- or bottom-coding by definition eliminates detailed inferences about the distribution beyond the thresholds, which is often where interest lies. Chopping tails also negatively impacts estimation of whole-data quantities. For example, Kennickell and Lane (2006) show that commonly used top codes badly distort estimates of the Gini coefficient, a key measure of income inequality.

Analysts can adjust inferences to correct for the effects of some forms of aggregation and collapsing, although for others (aggregation of geography, in particular) the analyst may not have any recourse but to make strong implicit assumptions about the nature of the heterogeneity at finer levels of aggregation/collapsing, e.g., that it is inconsequential. Analysts can correct for underestimation of variances caused by recoding by imputing the missing fine detail as part of a multiple imputation approach (Rubin 1987). For example, when ages are released as five-year intervals rather than exact values, analysts can impute exact ages within the intervals using an appropriate distribution for the study population; see Heitjan and Rubin (1990). For top-coding, analysts can treat the top-coded values as nonignorable missing data (e.g., values are missing because they are large), and use techniques from the missing data literature to adjust inferences (e.g., Jenkins et al. 2009). Like all nonignorable missing data problems, finding appropriate models is challenging, because the top-coded data contain no information on the top-coded values beyond the fact that they are large.

DATA SWAPPING

Data swapping has many variations, but all involve switching the data values for selected records with those for other records (Dalenius and Reiss 1982). For example, in the public-use files for the American Community Survey, the Census Bureau switches “a small number of records with similar records from

a neighboring area.”³ In the public-use files for decennial census, the Census Bureau swaps entire households across street blocks after matching on household size and a few other variables. The National Center for Education Statistics uses data swapping “to reduce the risk of data disclosure for individuals or institutions by exchanging values for an identifying variable or set of variables for cases that are similar on other characteristics” (Krenzke et al. 2006, p. 1). The protection afforded by swapping is based in large part on perception: agencies expect that intruders will be discouraged from linking to external files, since any apparent matches could be incorrect due to the swapping.

As the quotes above suggest, agencies generally do not reveal much about the swapping mechanisms to the public. They almost always keep secret the exact percentage of records that were involved in the swapping, as they worry that this information could be used to reverse-engineer the swapping.⁴ For related reasons, many agencies do not reveal the criteria that determine which records were swapped.

What are the impacts of swapping for secondary data analyses? The answer to this question is nuanced. Obviously, analyses not involving the swapped variables are completely unaffected. The marginal distributions (un-weighted) of all variables also are unaffected. However, analyses involving the swapped and un-swapped variables jointly can be distorted. Winkler (2007) shows that swapping 10 percent of records randomly essentially destroys the correlations between the swapped and un-swapped variables. Drechsler and Reiter (2010) show that even a 1-percent swap rate can cause actual coverage rates of confidence intervals to dip far below the nominal 95-percent rate. However, when swap rates are very small (less than 1 percent), it is reasonable to suppose that the overall impact on data quality is minimal; indeed, the error introduced by such small rates of swapping may well be swamped by the measurement error in the data themselves.

Without knowledge of the methodology used to implement the swapping—including the swap rate and the criteria used to select cases—secondary analysts can neither evaluate the impact of the disclosure treatment on inferences nor correct for any induced biases. In truth, even for analysts armed with the swap rate and the criteria, it is not trivial to incorporate the effects of swapping into inferences. For complex analyses involving swapped variables, the analyst would have to average over the unknown indicators of which records received swapping and the potential values of those records’ original data. Such an analysis has not appeared in the published literature.

3. Quote taken from the online documentation of the ACS at www.census.gov/acs/www/data_documentation/publicuse_microdata_sample/.

4. There has been no published empirical research quantifying the additional disclosure risks caused by revealing the swap rate or other broad details of the swapping mechanism.

ADDING NOISE

A different data perturbation strategy is to add random noise to sensitive or identifying data values (e.g., Brand 2002; Kim 1986). For example, in the public-use microdata for the Quarterly Workforce Indicators, the Census Bureau adds noise to continuous employment sizes and payrolls of establishments. The noise for each establishment is drawn randomly from a symmetric distribution with a hole in the middle, so that released values are guaranteed to be a minimum distance away from the actual values. This minimum distance is kept secret by the Census Bureau. The Census Bureau also adds noise to ages of individuals in large households in the public-use files of the decennial census and the American Community Survey. The variance of this noise distribution is kept secret by the Census Bureau. The UK Office of National Statistics (ONS) perturbs nominal variables like race and sex for sensitive cases in public-use samples from the demographic census using post-randomization techniques (Gouweleeuw et al. 1998). The basic idea is to change original values according to an agency-specified transition matrix of probabilities, e.g., switch white to black with probability 0.3, white to Asian with probability 0.2, etc. The change probabilities can be set so as to preserve the marginal frequencies from the confidential data. The ONS does not release the transition matrices in the public-use files, as it worries that doing so might decrease confidentiality protection (Bycroft and Merrett 2005).

Data perturbations can reduce disclosure risks because the released data are noisy, making it difficult for intruders to match on the released values with confidence. However, the noise also impacts the quality of secondary analyses. For numerical data, typically the noise is generated from a distribution with mean zero. Hence, point estimates of means remain unbiased, although the variance associated with those estimates may increase. Marginal distributions can be substantially altered, since the noise stretches the tails of the released variables. When using a noisy variable as a predictor in a regression model, analysts should expect attenuation in the regression coefficients, since the noise has the equivalent effect of introducing measurement error in the predictors (Fuller 1993). For categorical data, analysis of the released categories without any adjustments can lead to biased estimates for analyses not purposefully preserved by the perturbation.

Analysts can account for noise in inferences by treating the released data file as if it contains measurement errors (Fuller 1993; Little 1993). Techniques for handling measurement error exist for a variety of estimation tasks, including most regression models. However, the analyst must know the noise distribution to implement measurement error models effectively. Unfortunately, as indicated in the examples mentioned above, agencies usually do not release enough details about the noise infusion to enable appropriate adjustments. Hence, the analyst is left with little choice but to ignore the measurement errors and trust that their analysis has not been unduly compromised by the disclosure treatment.

There are versions of added noise that are guaranteed to have minimal impact on specific secondary inferences. For example, it is possible to add noise in ways that precisely preserve means and covariance matrices (Ting, Fienberg, and Trottini 2008; Shlomo and de Waal 2008), so that regression coefficients are perfectly preserved. The noise still may impact other analyses, the extent to which remains difficult to detect.

Partially Synthetic Data

Partially synthetic data comprise the units originally surveyed with some collected values, such as sensitive values at high risk of disclosure or values of key identifiers, replaced with multiple imputations. For example, suppose that the agency wants to replace income when it exceeds \$100,000—because the agency considers only these individuals to be at risk of disclosure—and is willing to release all other values as collected.⁵ The agency generates replacement values for the incomes over \$100,000 by randomly simulating from the distribution of income conditional on all other variables. To avoid bias, this distribution also must be conditional on income exceeding \$100,000. The distribution is estimated using the collected data and possibly other relevant information. The result is one synthetic data set. The agency repeats this process multiple times and releases the multiple data sets to the public. The multiple data sets enable secondary analysts to reflect the uncertainty from simulating replacement values in inferences.⁶

Among all the SDL techniques described here, partial synthesis is the newest and least commonly used. It has been employed mainly by government agencies. For example, the Federal Reserve Board in the Survey of Consumer Finances replaces monetary values at high disclosure risk with multiple imputations, releasing a mixture of these imputed values and the un-replaced, collected values (Kennickell 1997). The Census Bureau has released a partially synthetic public-use file for the Survey of Income and Program Participation that includes imputed values of Social Security benefits information and dozens of other highly sensitive variables (Abowd, Stinson, and Benedetto 2006). The Census Bureau protects the identities of people in group quarters (e.g., prisons, shelters) in the American Community Survey by replacing quasi-identifiers for records at

5. Alternatively, the agency could synthesize the key identifiers for the individuals with incomes exceeding \$100,000, with the goal of reducing risks that these individuals might be identified.

6. A stronger version of synthetic data is full synthesis, in which all values in the released data are simulated (Rubin 1993; Raghunathan, Reiter, and Rubin 2003; Reiter 2005b). No agencies have adopted the fully synthetic approach as of this writing, although agencies in Germany (Drechsler, Bender, and Rässler 2008) and New Zealand (Graham, Young, and Penny 2009) have initiated research programs into releasing fully synthetic products. This approach may become appealing in the future if confidentiality concerns grow to the point where no original data can be released in unrestricted public-use files.

high disclosure risk with imputations (Hawala 2008). The Census Bureau also has developed synthesized origin destination matrices, i.e., where people live and work, available to the public as maps via the Web (Machanavajjhala et al. 2008). Partially synthetic public-use data sets are in the development stage for the Longitudinal Business Database (Kinney and Reiter 2007) and the Longitudinal Employer-Household Dynamics Database (Abowd and Woodcock 2004). Other examples of applications include Little, Liu, and Raghunathan (2004), An and Little (2007), and An, Little, and McNally (2010).

Current practice for generating partially synthetic data is to employ sequential modeling strategies based on parametric or semi-parametric models akin to those for imputation of missing data (Raghunathan et al. 2001). The basic idea is to impute any confidential values of Y_1 from a regression of Y_1 on $(Y_2, Y_3, \text{etc.})$, impute confidential values of Y_2 from a regression of Y_2 on $(Y_1, Y_3, \text{etc.})$, impute confidential values of Y_3 from a regression of Y_3 on $(Y_1, Y_2, \text{etc.})$, and so on. Each regression model is tailored for the variable at hand, e.g., a logistic regression when Y_3 is binary. Several researchers have developed non-parametric and semi-parametric data synthesizers that can be used as conditional models in the sequential approach, for example, regression trees (Reiter 2005c), quantile regression (Huckett and Larsen 2007), kernel density regressions (Woodcock and Benedetto 2009), Bayesian networks (Young, Graham, and Penny 2009), and random forests (Caiola and Reiter 2010). (See Mitra and Reiter 2006 for a discussion of handling survey weights in partial synthesis.)

What are the impacts of partial synthesis for data analysis? When the replacement imputations are generated effectively, analysts can obtain valid inferences for a wide class of estimands via a simple process: (1) in each data set, estimate the quantity of interest and compute an estimate of its variance; and (2) combine the estimates across the data set using simple rules (Reiter 2003a; Reiter and Raghunathan 2007) akin to those for multiple imputation of missing data (Rubin 1987). These methods automatically propagate the uncertainty from the disclosure treatment, so that analysts can use standard methods of inference in each data set without the need for measurement error models. However, the validity of synthetic data inferences depends on the validity of the models used to generate the synthetic data. The extent of this dependence is driven by the amount of synthesis. If entire variables are simulated, analyses involving those variables reflect only those relationships included in the data generation models. When these models fail to reflect accurately certain relationships, analysts' inferences also will not reflect those relationships. Similarly, incorrect distributional assumptions built into the models will be passed on to the users' analyses. On the other hand, when only small fractions of data are simulated, inferences may be largely unaffected.

Agencies that release partially synthetic data often release information about the synthesis models, for example, the synthesis code (without parameter estimates) or descriptions of the modeling strategies. This can help analysts decide

whether or not the synthetic data are reliable for their analyses; for example, if the analyst is interested in a particular interaction effect that is not captured in the synthesis models, the synthetic data are not reliable. Still, it can be difficult for analysts to determine if and how their inference will be affected by the synthesis, particularly when nonparametric methods are used for data synthesis.

Concluding Remarks

From a secondary analyst's point of view, this article paints a perhaps distressing picture. All disclosure treatments degrade statistical analyses, and it is difficult if not impossible for analysts to determine how much one's analysis has been compromised. While it is theoretically feasible to adjust inferences for certain SDL techniques, agencies typically do not provide enough information to do so. The situation can be even more complicated in practice; for example, the ICPSR employs recoding, swapping, and perturbation simultaneously to protect some of its unrestricted access data. However, the picture is not as bleak as it may seem. First, many public-use data sets of probability samples have modest amounts of disclosure treatment (other than the often extensive geographic recoding), so that most inferences probably are not strongly affected.⁷

Second, there is ongoing research on developing ways to provide automated feedback to users about the quality of inferences from the protected data. One idea is verification servers (Reiter, Oganian, and Karr 2009), which operate as follows. The data user performs an analysis of the protected data, and then submits a description of the analysis to the verification server; for example, regress attribute 5 on attributes 1, 2, 4, and the square root of attribute 6. The verification server performs the analysis on both the confidential and protected data and, from the results, calculates analysis-specific measures of the fidelity of the one to the other. For example, for any regression coefficient, measure the overlap in its confidence intervals computed from the confidential and protected data. The verification server returns the value of the fidelity measure to the user. If the user feels that the intervals overlap adequately, the altered data have high utility for their analysis. With such feedback, analysts can avoid publishing—in the broad sense—results with poor quality, and be confident about results with good quality. Verification servers have not yet been deployed. A main reason is that fidelity measures provide intruders with information about the real data, albeit in a convoluted form, that could be used for disclosure attacks. Finding ways to reduce the risks of providing fidelity measures are topics of ongoing research.

7. For many data sets, agencies believe that the act of random sampling provides strong protection itself. This is true provided that (1) intruders do not know who participated in the survey; and (2) data subjects are not unique in the population on characteristics known by intruders.

Third, agencies are actively developing methods for providing other forms of access to confidential data. Several agencies are developing remote analysis servers that report the results of analyses of the underlying data to secondary data users without allowing them to see the actual microdata. Such systems can take a variety of forms. For example, the system might allow users to submit queries for the results of regressions, and the system reports the estimated coefficients and standard errors. This type of system is in use by the National Center for Education Statistics, and similar systems are being developed by the Census Bureau and National Center for Health Statistics.

One might ask why analysis servers alone are not adequate for users' needs. These systems have two primary weaknesses. First, users have to specify the analysis without seeing the data and with limited abilities to check the appropriateness of any model assumptions. This is because exploratory graphics and model diagnostics, like residual plots, leak information about the original data. It may be possible to disclosure-proof the diagnostic measures to deal with this problem (Sparks et al. 2008); for example, the Census Bureau's remote access system will provide simulated regression diagnostics (Reiter 2003b). Second, with judicious queries an analyst might be able to determine actual values in the data, e.g., learn a sensitive outcome by estimating a regression of that outcome on an indicator variable that equals one for some unique value of a covariate and zero otherwise (Gomatam et al. 2005).⁸ Hence, analysis servers have to be limited in terms of what queries they respond to. Alternatively, the server might add noise to the outputs (Dwork 2006), although this is challenging to implement in a way that ensures confidentiality protection.

What does the future hold for access to confidential data in the social sciences? We can only speculate, but concerns over data confidentiality are becoming acute enough that some organizations in the near future may seek to avoid making any original microdata available as unrestricted access files. Existing methods of SDL are not likely to offer solutions: as resources available to intruders expand, the alterations needed to protect data with traditional techniques like recoding, swapping, and adding noise may become so extreme that, for many analyses, the altered data are practically useless. In this world, the future may resemble the form outlined by the National Research Council (2005, 2007), which suggested a tiered approach to data access involving multiple modalities. For highly trusted users with no interest in breaking confidentiality and who are willing to take an oath not to do so, agencies can provide access to data via licensing or virtual data enclaves. For other users, agencies can provide access to strongly protected microdata—which probably need to be fully synthetic data sets if intense alteration is necessary to ensure adequate

8. Verification servers may be more immune to such attacks than remote analysis servers. The verification server only needs to report a measure of analysis quality, which can be coarsened and still useful. The analysis server needs to report results from the actual analysis without perturbations; otherwise, it may not yield enough value added to the released microdata.

protection—coupled ideally with a verification server to help users evaluate their results. For many data users, the protected microdata would be adequate for their purposes; others could use the protected microdata to gain familiarity with the data, e.g., to roughly gauge whether the data can answer the question of interest or assess the need for transformations of data, before going through the costs of applying for restricted access or working in a secure data center. Such tiered access would allow trusted researchers access to high-quality data when they are willing to accept the costs to do so, and enable others in the public to obtain reasonable answers for many questions without meeting the high hurdles of restricted access.

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